Lecture on

Waves & Oscillations

Classification of Waves

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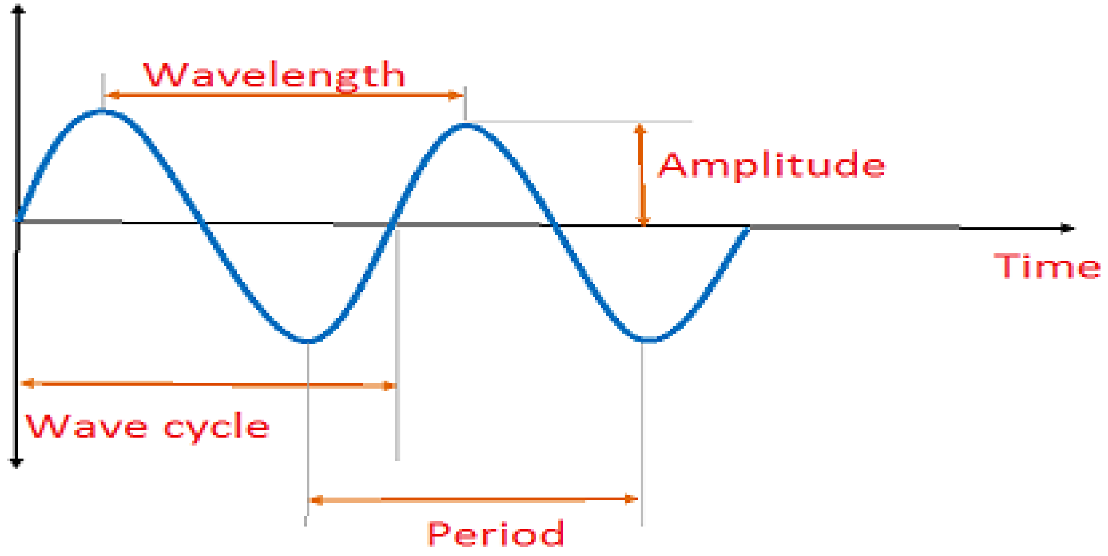
Classification of Waves

Wave: **A wave is a travelling disturbance** that transports energy from place to place.

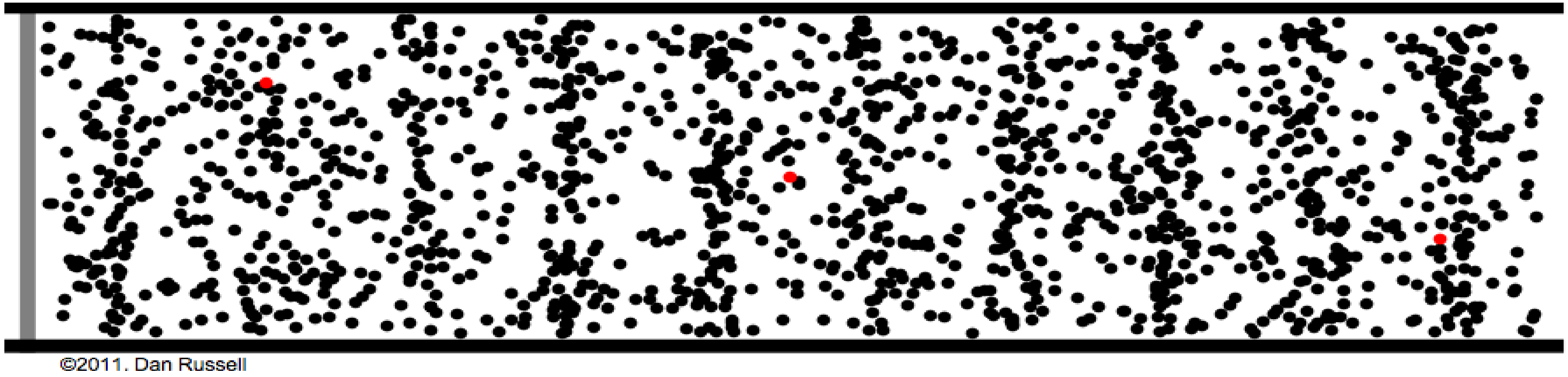
Wave is an energy transfer phenomena in which particle position does not change.

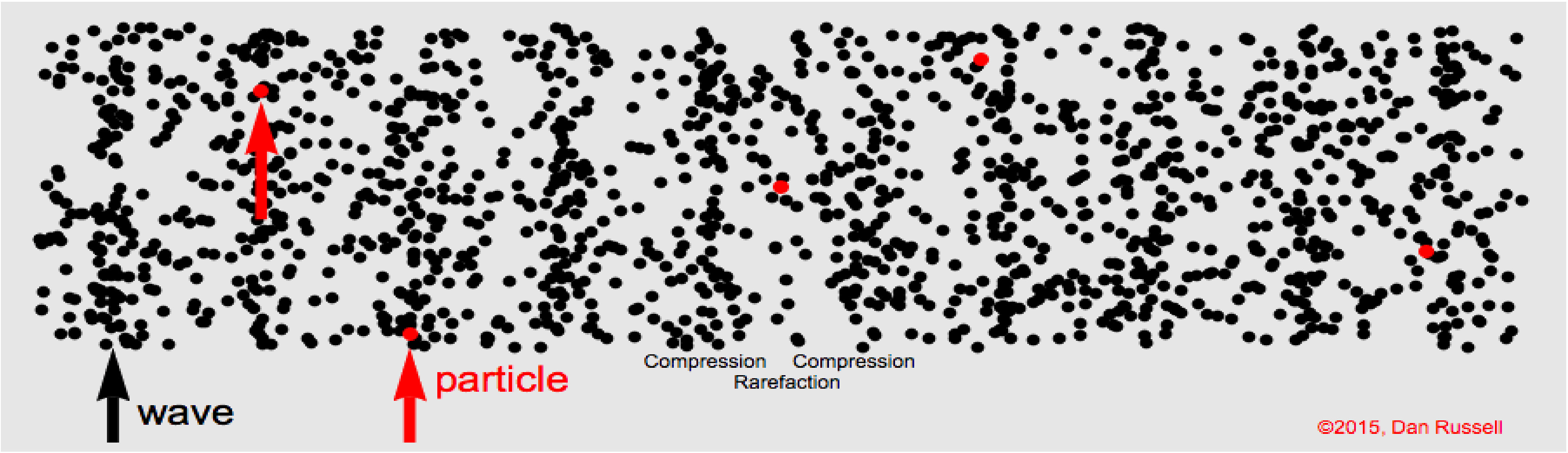


Classification of Waves

So that, Wave is an energy transfer phenomena in which particle position effectively does not change from equilibrium.

The information can also be transferred by wave.





Classification of Waves

Different types of waves have a different set of characteristics. Based on the orientation of particle motion and direction of energy, there are mainly three categories of waves:

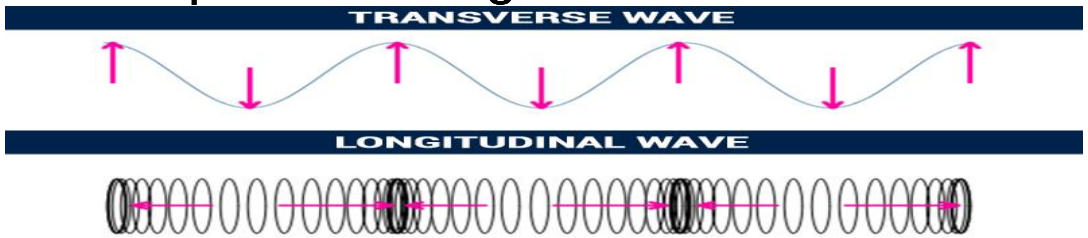
- ❑ 1. Mechanical Waves
- ❑ 2. Electromagnetic Waves
- ❑ 3. Matter Waves

Mechanical Waves: A mechanical wave is a wave that is an oscillation of matter and is responsible for the transfer of energy through a medium. Example: Sound wave, water wave, etc.

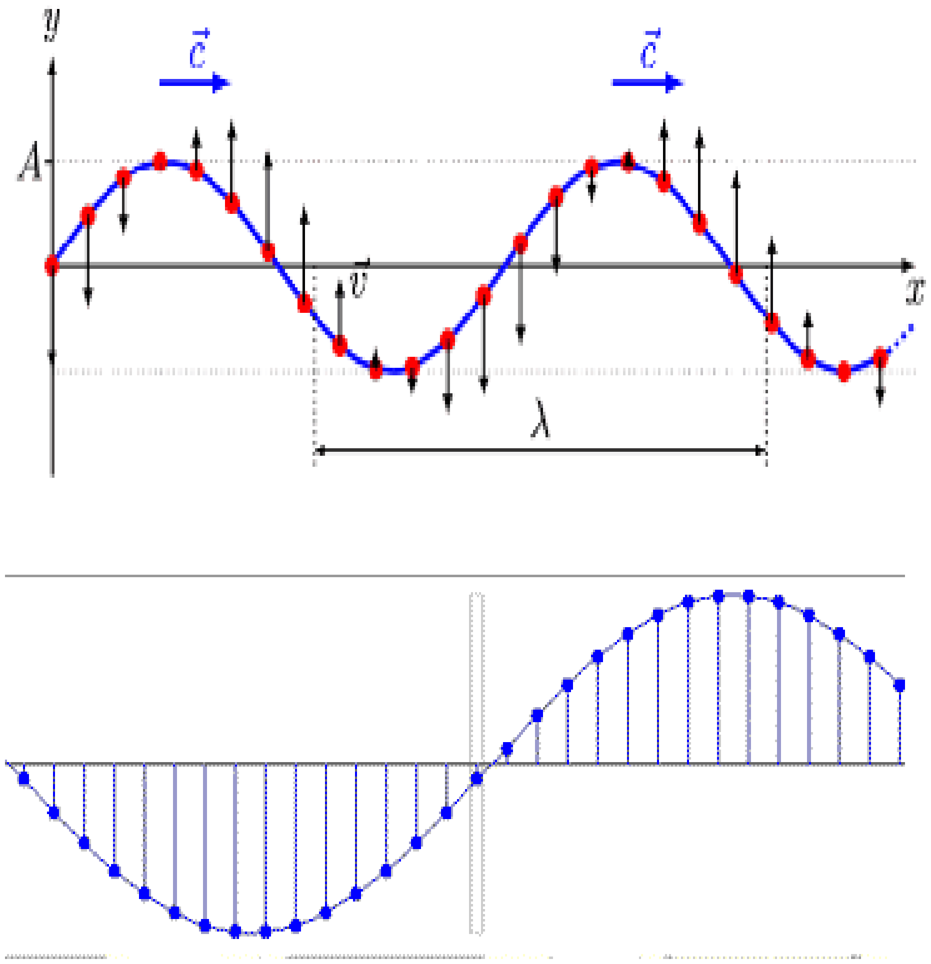
There are two types of mechanical waves: (i) **Longitudinal waves (ii) Transverse waves.**

EM Waves: Electromagnetic waves are created by a fusion of electric and magnetic fields. Unlike mechanical waves electromagnetic waves do not need a medium to travel. Example: Light wave, Microwave, etc.

Matter Wave: The dual nature of matter; its ability to exist both as a particle and a wave is called matter wave. Example: De-Broglie wave, beam of electrons, etc.



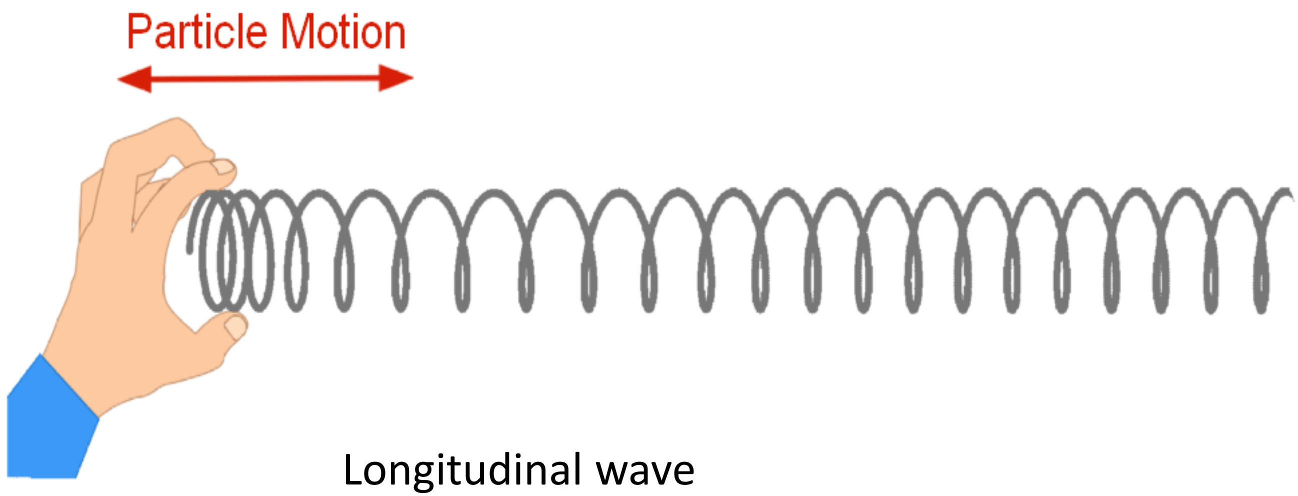
Classification of Waves



Transverse wave

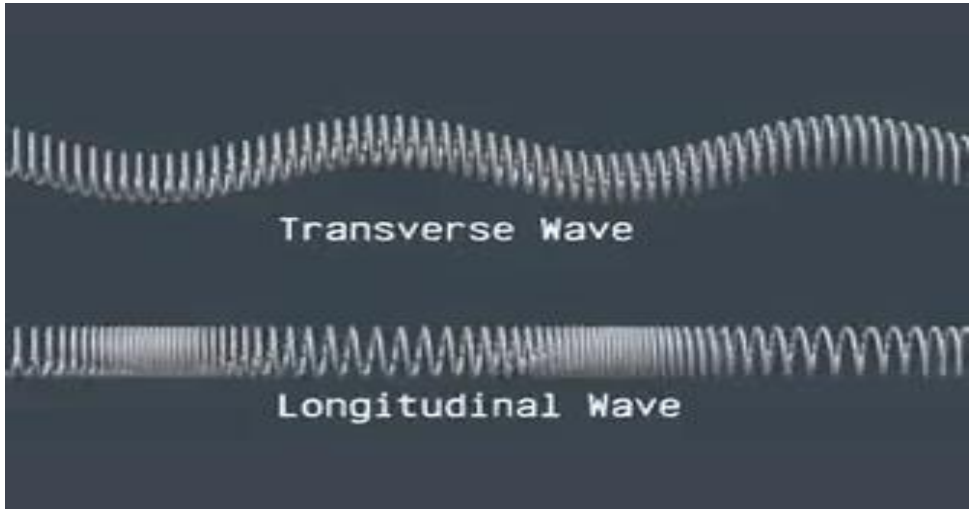
The motion of particles in a wave can be either **perpendicular** to the wave direction (**transverse wave**) or
parallel to the wave direction (**longitudinal wave**).

Classification of Waves



Longitudinal wave

The motion of particles in a wave can be either **perpendicular** to the wave direction (**transverse wave**) or **parallel** to the wave direction (**longitudinal wave**).



Comparison between Transverse wave and Longitudinal wave

Classification of Waves

Electromagnetic Wave consists of electric field and magnetic field, which propagates through space and carry momentum and electromagnetic radiant energy. Example: X-rays, light wave, etc.

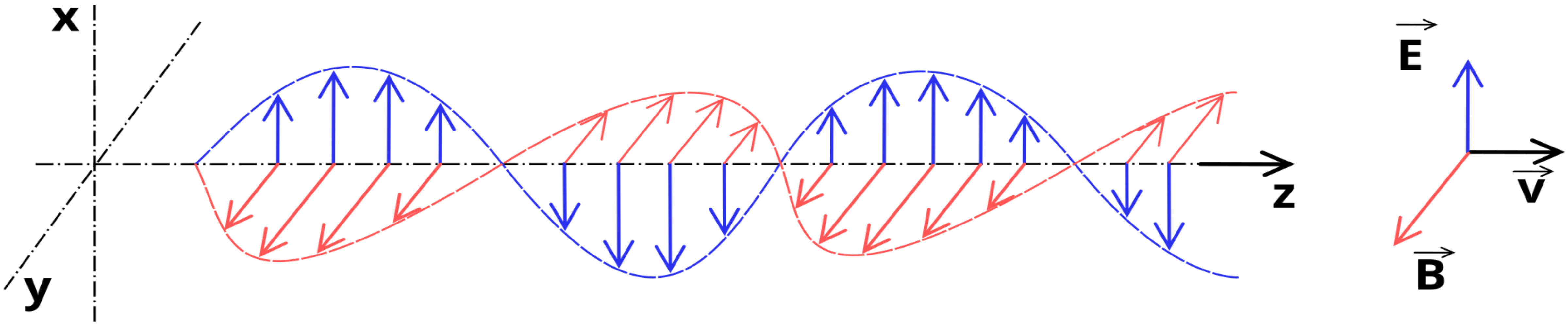


Fig: A sinusoidal EM wave propagating along the positive z-axis, showing the Electric field (blue) and magnetic field (red) vectors.

Electromagnetic radiation is one of the ways that energy travels through space. EM waves are Transverse waves. James Clerk Maxwell first invented (In 1862–64) the EM waves and established 4 ‘’Maxwell’s EM Equation/ ’Maxwell’s Equation’’ for EM Wave theory in Electrodynamics and later Maxwell's equations were confirmed by Heinrich Hertz (in 1887) through experiments with radio waves.

Electromagnetic radiation is commonly referred to as "light", EM, EMR, or Electromagnetic waves.

Classification of Waves

Given relation is $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$\Rightarrow \mu_0 = \frac{1}{\epsilon_0 c^2}$

Electromagnetic Wave: Maxwell's Equation

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

$$\boxed{\begin{aligned} &\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} && \text{Faraday} \\ &\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} && \text{Ampere} \\ &\left.\begin{aligned} &\nabla \cdot \mathbf{D} = \rho \\ &\nabla \cdot \mathbf{B} = 0 \end{aligned}\right\} && \text{Gauss} \end{aligned}}$$

$$D = \varepsilon_0 E + P \qquad \mathbf{D} \equiv \varepsilon_0 \mathbf{E}$$

$c = \frac{1}{\sqrt{\mu\varepsilon}}$ (speed of light)

In Free Space (Vacuum):

$\mu_0 = 4\pi \cdot 10^{-7}$ [H/m]

$\varepsilon_0 = 8.854 \cdot 10^{-12}$ [F/m]

$c_0 = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ =299,795,638 [m/s]

E is the electric field, **B** the magnetic field, ρ the electric charge density, **J** the current density, ε_o is the vacuum permittivity, μ_0 is the vacuum permeability, **H** is the magnetic field intensity, **P** is the polarization density, and **D** is the electric displacement field.

$\nabla . \boldsymbol{D} = \rho$ → Gauss's Law for Electric Field

$\nabla . \boldsymbol{B} = 0$ → Gauss's Law for Magnetic Field

$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$ → Faraday's Law for Electric Field

$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}$ → Ampere's Law for Magnetic Field

Name	Equation	
	Integral form	Differential form
Faraday's law of induction	$\oint_c \vec{E} \cdot d\vec{l} = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
Ampère-Maxwell law	$\oint_c \vec{H} \cdot d\vec{l} = \iint_S \vec{J} \cdot d\vec{S} + \iint_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
Gauss' electric law	$\oint\!\!\oint_S \vec{D} \cdot d\vec{S} = \iiint_V \rho \, dV$	$\nabla \cdot \vec{D} = \rho$
Gauss' magnetic law	$\oint\!\!\oint_S \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$

Classification of Waves

Matter-wave is an important part of quantum physics, being half of wave-particle duality. ***At all scales where measurements have been practical, matter exhibits wave-like behavior***. For example, a beam of electrons can be diffracted just like a beam of light or a water wave. **Matter waves are also known as de Broglie waves**.

The *de Broglie wavelength* is the wavelength, λ, associated with a particle with momentum p through the Planck constant, h:

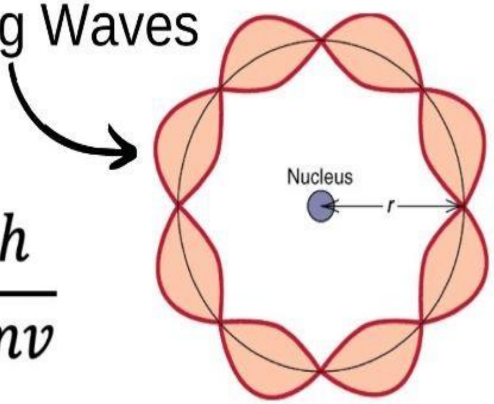
$$\lambda = \frac{h}{p}$$

$$Momentum\ P = mv = \frac{E}{c} = \frac{h\upsilon}{c} = \frac{h}{\lambda}$$

Wave-like behavior of matter has been experimentally demonstrated, first for electrons in 1927 and for other elementary particles, neutral atoms and molecules in the years since.

Matter wave concepts are widely used in the study of materials where different wavelength and interaction characteristics of electrons, neutrons, and atoms are leveraged for advanced microscopy and diffraction technologies.

Standing Waves



$$\lambda = \frac{h}{mv}$$

The modified momentum, $\mathbf{p} = \gamma m_0 \mathbf{v}$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

γ *is the relativistic factor or Lorentz factor*

Quantum mechanics

$$i\hbar\frac{d}{dt}|\Psi\rangle = \hat{H}|\Psi\rangle$$

Schrödinger equation

Classification of Waves

Thank You

The END